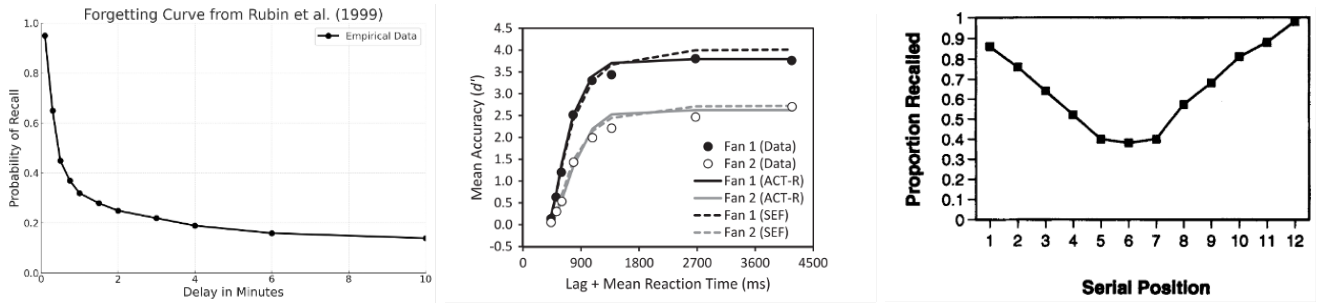

(a) Human memory shows rapid early decay followed by stabilization over time [40].

(b) Accuracy of human memory declines as item associations increase. Fan 1 outperforms Fan 2, especially at longer lags [46].

(c) Human memory exhibits a U-shaped serial position curve with primacy and recency effects (positional bias) [38].

Figure 3: Comparison of human memory patterns with LLM behavior. (a) LLMs show reduced recall accuracy with increasing list length, consistent with human time-interference patterns. (b) LLM recall accuracy declines under increased associations to an item, reflecting human-like memory patterns, with both LLMs and humans exhibiting the fan effect. (c) LLMs exhibit no sensitivity to serial position effects and maintain robust performance across positions, in contrast to humans who show clear primacy and recency effects.

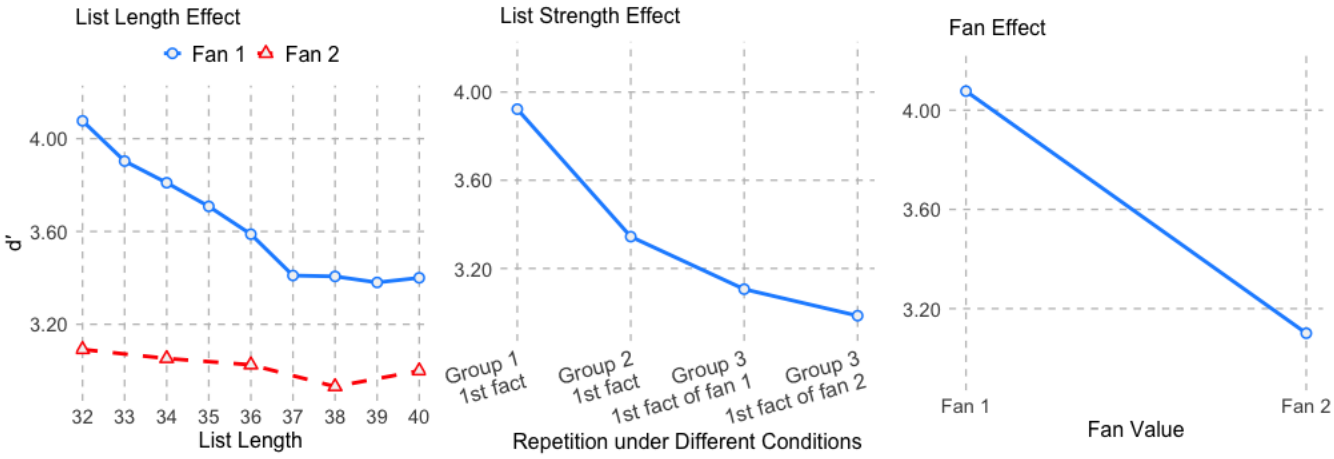

Figure 4: LLM performance across memory effects measured by d'. Mean *d'* values are shown for three experimental groups: (a) memory sensitivity and discriminability decrease as list length increases, particularly under lower fan conditions; (b) LLM shows that repetition progressively reduces discriminability; and (c) fan effect, demonstrating lower overall *d'* for Fan 2 compared with Fan 1, consistent with classical interference patterns in human associative recognition.

interference. The second subfigure illustrates the list strength effect, showing a trend consistent with the recognition accuracy patterns observed in our LLM experiments. The mean *d'* values in LLMs exceed those in human participants, indicating that the model demonstrates greater discriminative capacity under different list strength conditions.

5 Materials and Methods

All experiments in this study are mainly conducted using an online large language model, specifically the GPT-4 architecture. More precisely, we utilize the `'gpt-4-0125-preview'` variant provided through the OpenAI API. The subsequent sections will systematically detail the evaluation metrics, procedures, and designs of experiments.

Evaluation Metrics

Performance was assessed by recording recall accuracy, false alarm rate, hit rate, and standard deviation. According to the confusion matrix presented in Table 3, these evaluation metrics are defined mathematically as follows:

$$\text{RECALL ACCURACY} = \frac{TP + TN}{TP + TN + FN + FP} \tag{2}$$

	Predicted Positive	Predicted Negative
Actual Positive	TP	FN
Actual Negative	FP	TN

Table 3: Confusion Matrix in Memory Experiment

$$\text{FALSE-ALARM RATE} = \frac{FP}{FP + TN} \tag{3}$$

$$\text{HIT RATE} = \frac{TP}{TP + FN} \tag{4}$$

Experimental Setup

Dataset 1: "Person Location Lists": We tasked large language models with memorizing facts during a study phase. Subsequently, a recognition task was performed during the test phase, where the language models were told to distinguish between targets (studied facts) and foils (non-studied facts, which were rearranged items derived from the studied facts). All studied facts and foils were presented in question format in the test phase.

We interacted with the language models using task-specific prompts (Appendix Table S3). The format and content of these prompts were important for performing the recognition memory task accurately. We determined the final prompt design by various approaches, including interacting with the language models to get recommendations for optimal prompt structure for a recognition memory task. The outputs from the language models were then collected to assess recognition performance. Ideally, the optimal outcome would be `'yes'` for the first half of the questions and `'no'` for the second half. The final evaluation results were obtained by comparing the recognition results with the ground truth.

Dataset 2:"Target words and Associates Lists": In this experiment, language models were told to act as participants in a memory experiment and were first tasked with memorizing a study list of words during the study phase. In the subsequent test phase, participants were presented with a recognition task. We engaged the language models using the example prompt from the Appendix Table S4. The method for generating the prompt was similar to that used in Dataset 1.

Experiment 1: List-Length Effect

Design: The first four experiments in this study were constructed upon a shared framework of study lists composed of person–location pairs. Within this framework, we define three experimental groups according to their fan values: **Group 1** (fan value = 1): consisted of 32 unique and distinct person–location pairs. **Group 2** (fan value = 2): consisted of 16 distinct persons and 16 distinct locations, arranged so that each person and each location appears twice, yielding 32 person–location pairs in total. **Group 3** (integration of fan values 1 and 2): consisted of 16 person–location pairs with a fan value of 1 and 16 pairs with a fan value of 2, derived from 24 distinct persons and locations (16 + 8). Detailedly, Group 3 included 16 person-location pairs for a fan value of 1 and 16 person-location pairs for a fan value of 2 derived from arrangements of 8 distinct persons and locations, ensuring the robustness of results. This common framework serves as the basis for all subsequent manipulations.

This experiment investigating the list length effect was designed by building upon the first two groups. As the number of items in a list increases, the likelihood of correctly recognizing an item from that list typically decreases. Therefore, we varied the length of the study list within the group to explore the impact of list length. The initial list length remained consistent with the fan effect experiment, containing 32 person-location pairs.

For Group 1, we examined list lengths ranging from 32 to 40, incrementing by one study fact each time (32, 33, 34, ..., 40). The study list included 40 person-location pairs with a fan value of 1, derived from lists of person and location words. For Group 2, the experimental design was consistent, with the only difference being that list lengths were incremented by two study facts at each step.

Experiment 2: List-Strength Effect

Design: The experiments investigating the list strength effect were also based on the three groups from the fan effect study. The "strength" of an item in memory refers to how well it is encoded or how robustly it is represented, and strength can be increased by repeated exposure to the item. Therefore, within each group, we repeated a specific study list sentence multiple times.

To streamline the experimental design, Group 1 and Group 2 were subjected to five repetitions of the first study fact. In Group 3, we conducted two distinct experiments: the first involved repeating the first study fact with a fan value of 1 five times, followed by recording the results; the second replicated this process with a fan value of 2, ensuring consistent data collection across conditions.

Experiment 3: Fan Effect

Design: In our investigation of fan effects using the controlled variable method, it is essential to maintain a constant study list size. Additionally, fixing the study list size inherently fixes the question list size, which is twice the length of the study list. In the fan effects experiments, we explored the impact of different fan levels and their integration. We standardized the study list length at 32 person-location pairs. There are 2 groups in this experiment, already introduced in Experiment 1, which examine a fan value of 1 only and a fan value of 2 only.

Experiment 4: Nonsense Effect

Design: The "nonsense effect" refers to the phenomenon where participants are more likely to incorrectly recognize or falsely recall something because it appears meaningless or out of context. In the nonsense effect experiment, we focused exclusively on Group 1. Our experiment consists of three sections: one involving only the replacement of persons, another focusing solely on the replacement of locations, and the third replacing both persons and locations simultaneously. For all replaced words in Group 1, we replaced them with nonsensical words consisting of random combinations of letters and numbers, while maintaining the original word length. For example, "`library`" might be randomly replaced with "`s3mc01m`". It is important to note that we replaced all words in both the study list and the question list.

Experiment 5: Position Effect

Design: For the position effect, we will provide the language model with the first 6 lists, each containing the first 12 associated words. After the language model memorizes each study list, we will immediately give it a recognition test list to recall. Each recognition test list consists of 42 words, including 12 studied items and 30 non-studied items. The non-studied items are divided into three types: (a) 6 critical lure words, (b) 12 words generally unrelated to any items on the 6 lists, and (c) 12 words weakly related to the lists (2 per list in serial positions 14 and 15). We will obtain the recall rate of the studied words in the test list.

Experiment 6 and 7: DRM Effect and generalization ability

Design: This experiment is designed similarly to Roediger's experiment. We trained the language model to memorize a study list composed of words from 8 associate lists ($8 \times 15 = 120$ words in total). Following this, the language model was asked to perform a recognition test consisting of 48 words, including 24 studied words and 24 non-studied words. The 24 studied words were obtained by selecting 3 items from each of the 8 presented lists (always those in serial positions 1, 7, and 15). The non-studied words include 12 critical lures from all 12 lists (8 studied and 4 non-studied) and 12 items from the 4 non-studied lists (again, from serial positions 1, 7, and 15). If the language model recognized that a word in the recognition test appeared in the study list, it should be marked as "`old`"; otherwise, it should be marked as "`new`".

To further investigate whether immediate recall has a greater impact, we introduce two subgroup conditions: Immediate Recall Condition: The language model takes the recognition test immediately after memorizing the study list. Delayed Recall Condition: After memorizing the study list, the language model completes a series of simple math problems before taking the recognition test.

We measure the DRM effect by analyzing the false alarm rate and hit rate. Additionally, we obtain the recall rate of critical lure words (those from the studied lists) to represent the generalization ability. If the language model incorrectly recalls these critical lures, it demonstrates the model's ability to generalize these critical words based on their associates, even though they were not presented in the study list.

6 Discussion

The effects we examine here capture core aspects of memory performance. Some focus on associations, while others highlight how order and repetition shape recall. Psychologists studied these effects because each addresses a foundational question: whether memory is limited by capacity (list length), by interference (list strength, fan effect), by temporal order (position effect), by meaning (nonsense effect), by its reconstructive rather than reproductive nature (false memories), or by the tradeoff between preserving detail and enabling abstraction (generalization). Together, these paradigms show the tradeoffs that make memory both adaptive and fallible and explain why they remain benchmarks for understanding human and artificial systems. Our findings show that LLMs echo these tradeoffs, though with different balances, revealing both parallels and divergences in how natural and artificial systems remember. Brainerd, Reyna, and Ceci show how developmental shifts in false memory reflect the balance between verbatim and gist memory, pointing to next steps for testing whether artificial systems are also susceptible to these tradeoffs [13].

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